

Calc III Semester Outline & Review Topics

Chapter 12: 3 Dimensional Systems & Vectors

- Add, subtract, and multiply (dot and cross product) vectors and make vectors unit.
- Find equations of lines (given points, planes, parallel or perpendicular lines etc.).
- Find equations of planes (given points, planes, parallel or perpendicular lines etc.).
- Use the dot product formula (to find angles, projections, etc.).
- Determine if lines are intersecting, skew, or parallel.

Chapter 13: Vector Functions

- Determine the domain of a vector function.
- Find the limit of a vector function.
- Find the tangent vector (or tangent line) to a space curve at a point.
- Find the integral of vector functions.
- Given a vector function, find the arc length of the corresponding space curve.
- Reparametrize a vector function in terms of arc length.
- Find the curvature of a vector function.
- Find the TNB-Frames (unit tangent, unit normal, unit bi-normal) of a vector function.

Chapter 14: Partial Derivatives

- Determine if a limit of a point on a surface exists or does not exist.
- Find the partial derivative of any differentiable surface.
- Find second order (and higher order) partial derivatives.
- Solve chain rule problems for functions given dependent, intermediate, and independent variables (draw tree diagrams).
- Find the directional derivative of a surface at a point.
- Find tangent lines to a surface at a point in a direction.
- Find the gradient vector and use it to find the slope of steepest ascent.
- Find the equation of tangent planes and normal lines to a surface at a point.
- Given a surface, find local maximums, minimums, and saddle point, and their values.
- Given a surface find extreme maximums and minimums.

Chapter 15: Multiple Integrals

- Estimate the volume under a surface bounded by rectangular regions.
- Evaluate double integrals over general regions.

- Reverse the order of integration of double (and triple) integrals.
- Use polar coordinates to evaluate double integrals.
- Find the average value of a double (and triple) integral.
- Evaluate triple integrals to find the mass of a region given a mass density function.
- Evaluate triple integrals by converting to cylindrical coordinates.
- Evaluate triple integrals by converting to spherical coordinates.
- Find the Jacobian of a transformation of variables.
- Change the variables of a multiple integral using the Jacobian, give a transformation.

Chapter 16: Vector Fields

- Visualize and draw a given vector field.
- Find the divergence and curl of a point on a given vector field.
- Evaluate the line integral given a parameterized (or not) curve.
- Determine if a vector field is conservative.
- Use Green's Theorem to evaluate line integrals.